HW10 Warm-up: Technical Prototype Gate

Exposure Triangle — final low-fidelity iteration before implementation

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1. Prototype selection

- Continuing with **The Exposure Triangle** — an interactive lesson covering Camera Iris (aperture), Capture Duration (shutter speed), and Sensor Sensitivity (ISO).
- Approved in HW9 after two user tests validated the vocabulary rewrite and the bucket metaphor across both technically-exposed and non-technical profiles.

2. Final low-fidelity iteration

A round of iterations was applied before the HW10 Flask implementation. The iterations land in the prototype slide deck directly; the submitted slide PDF is exported from that state.

2.1 Vocabulary correction: Capture Window → Capture Duration

- "Window" risks reading as spatial framing rather than temporal length; "Duration" binds cleanly to the technical term *shutter speed*.
- "Fast shutter speed" ↔ "short capture duration" is a direct translation — matters for the bonus-path vocabulary bridge (§2.5) where scalar labels must map unambiguously to technical vocabulary.
- Consistent with the HW9 User Test Report's note that *"the substituted terms should be understood as working drafts, not final labels."*

2.2 New framing slide: "Press the Shutter" + why the UI is dark

Feedback flagged the dark background as unexplained. The new slide does two things at once:

- **Dramatic narrative of a single shutter press** — iris snapping to its chosen diameter, shutter lifting, light raining into a grid of millions of tiny buckets on the sensor, curtain falling — all in milliseconds. Serves double duty as a high-level "what does a camera do" orientation before the detailed teaching arc, and reinforces the bucket metaphor running through the lesson.
- **Why the page is dark** — Lightroom, Premiere, DaVinci Resolve, the photographer's darkroom all use dark chrome because the viewer's iris opens wider in dim rooms, making the visual system more sensitive to subtle differences in tone. Same reason movie theaters are black. Slide closes with *"Your eyes are doing photography on this page too."*

2.3 New interaction type: image comparison slider

- HW9 Sensor Sensitivity was a three-state discrete toggle (low / medium / high). Direct A/B comparison was missing.
- New comparison-slider slide: draggable divider between the low-sensitivity and high-sensitivity renders; drag reveals more of one or the other.
- Adds a new UI primitive (comparison slider) to the prototype's vocabulary of interactions — a response to a design challenge identified during planning: *"the slide format really constrains you."*
- Widening the range of interaction types — and establishing a reusable interaction library — answers that constraint better than making a single primitive carry all the pedagogical weight.

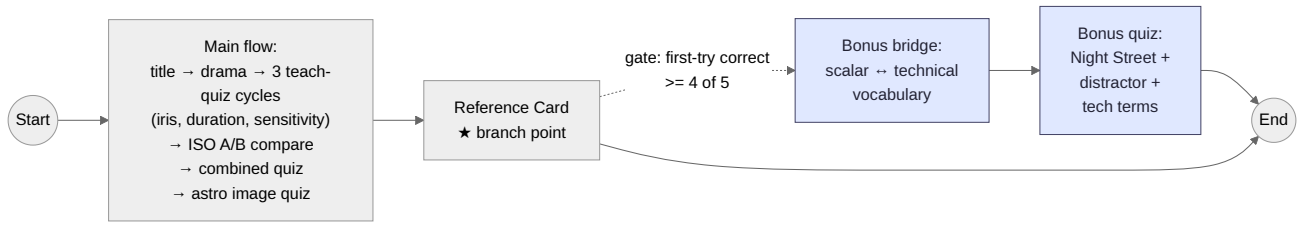
2.4 Quiz reorder

- Main-flow quizzes reordered so the image-choice astrophotography quiz is now last in the main flow (immediately before the reference card).
- Text-only night-street quiz moved off main flow and onto the bonus path (§2.5).
- Main flow now ends with a visual-reasoning checkpoint rather than a text one.

2.5 Branching: adaptive-difficulty path with distractor and technical vocabulary

A new terminal branching node implements a low-fidelity version of adaptive difficulty.

Flow diagram:



Properties the design enforces:

- **Single branch point, not cascading.** Fork happens only at the reference card, between the end of the main flow and termination. No quiz branches to another quiz-level branch. Avoids combinatorial explosion.
- **Terminal fork.** The bonus node has no further branches. Both paths rejoin at the same end state; only visible distinction is a subtle *"Bonus completed ✓"* marker on the reference card if the bonus was attempted.
- **Gated reveal, not forced.** Bonus is revealed only when at least four of the five main-flow multiple-choice quizzes were answered correctly on the *first* attempt. Second attempts and wrong answers are not punished — they simply end at the reference card as a normal completion.

The bonus path itself has two slides:

- **Vocabulary bridge** — a rosetta-stone table mapping each scalar label (wide/closed Camera Iris, short/long Capture Duration, low/high Sensor Sensitivity) to its technical term (aperture/f-stop, shutter speed, ISO). On-ramp from lesson vocabulary to the technical vocabulary the bonus quiz uses.
- **Bonus quiz** — a harder night-street scenario stacking three difficulty axes:

- **Technical vocabulary** (f/1.8, 1/500s, ISO 3200) instead of scalar terms.
- **Plausible distractor** — option E is "wide aperture, *moderate* shutter (1/60s), high ISO." Looks right but fails because 1/60s is too slow to freeze plate-spinning motion. Structural distractor: matches two of three correct choices, so rule-of-thumb pattern-matching does not solve it.
- **Irrelevant scenario detail** — wet pavement, sodium streetlamps — tests whether scene mood can be distinguished from exposure-relevant variables.

An inline design note on the bonus slide surfaces the fact that the node is a branching path implementing adaptive difficulty — so the branch is a documented design element readable from the slide itself, not a hidden state.

2.6 Second-chance quiz mechanic

- All multiple-choice quizzes now accept two attempts.
- **First wrong click:** the chosen option is grayed out (neutral disabled state), amber feedback text invites another try, other options stay interactive.
- **Second wrong click:** that option turns red, correct answer is revealed, the group locks.
- **Correct click at any point** advances the flow.

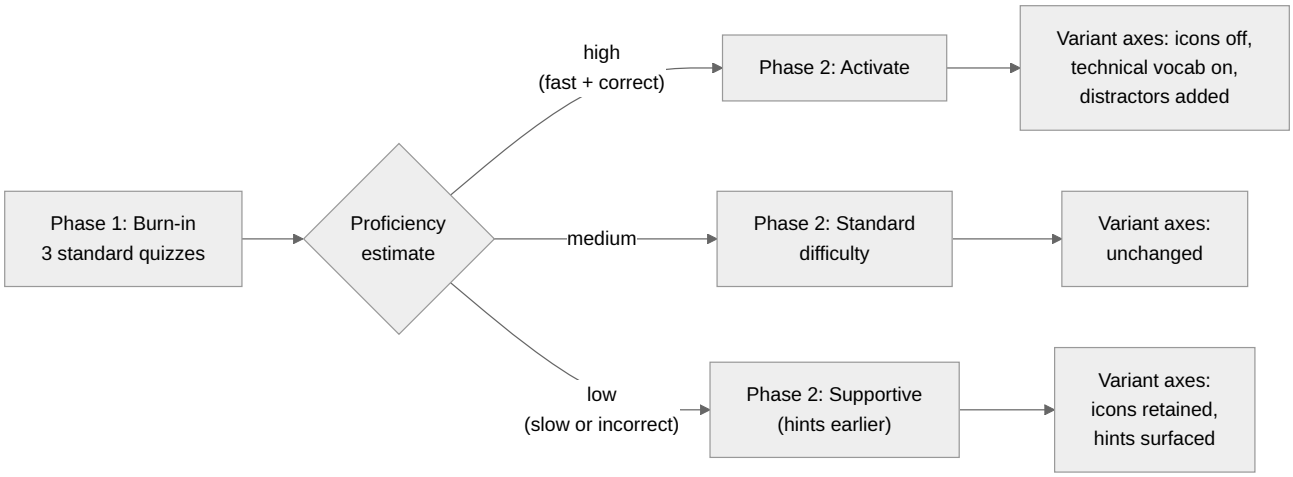
Rationale:

- Strongly subtractive/punitive error states can reduce motivation on first exposure to a concept, compared to neutral "try again" framings.
- Two-attempt interaction preserves the pedagogical value of being wrong (the user had to commit to an answer) without making a single misclick an unrecoverable failure.
- The gate in §2.5 tracks *first-try* correctness specifically, so second-attempt successes still count as learning but do not unlock the harder path.

3. Rationale document: adaptive-difficulty full design

The branching node in §2.5 is a low-fidelity stand-in for a more elaborate adaptive-difficulty system. The full version only makes sense to build once the Flask backend exists (HW10 main).

3.1 Three-phase structure



- **Burn-in:** first three multiple-choice quizzes are fixed and identical for all users. Latency and first-try correctness recorded.
- **Proficiency estimate:** simple heuristic combining (first-try correct count) and (response time). Not a model.
- **Activate:** from the fourth quiz on, content is drawn from pre-authored variants along difficulty axes. Proficiency estimate updates after each question; the system can escalate or de-escalate.

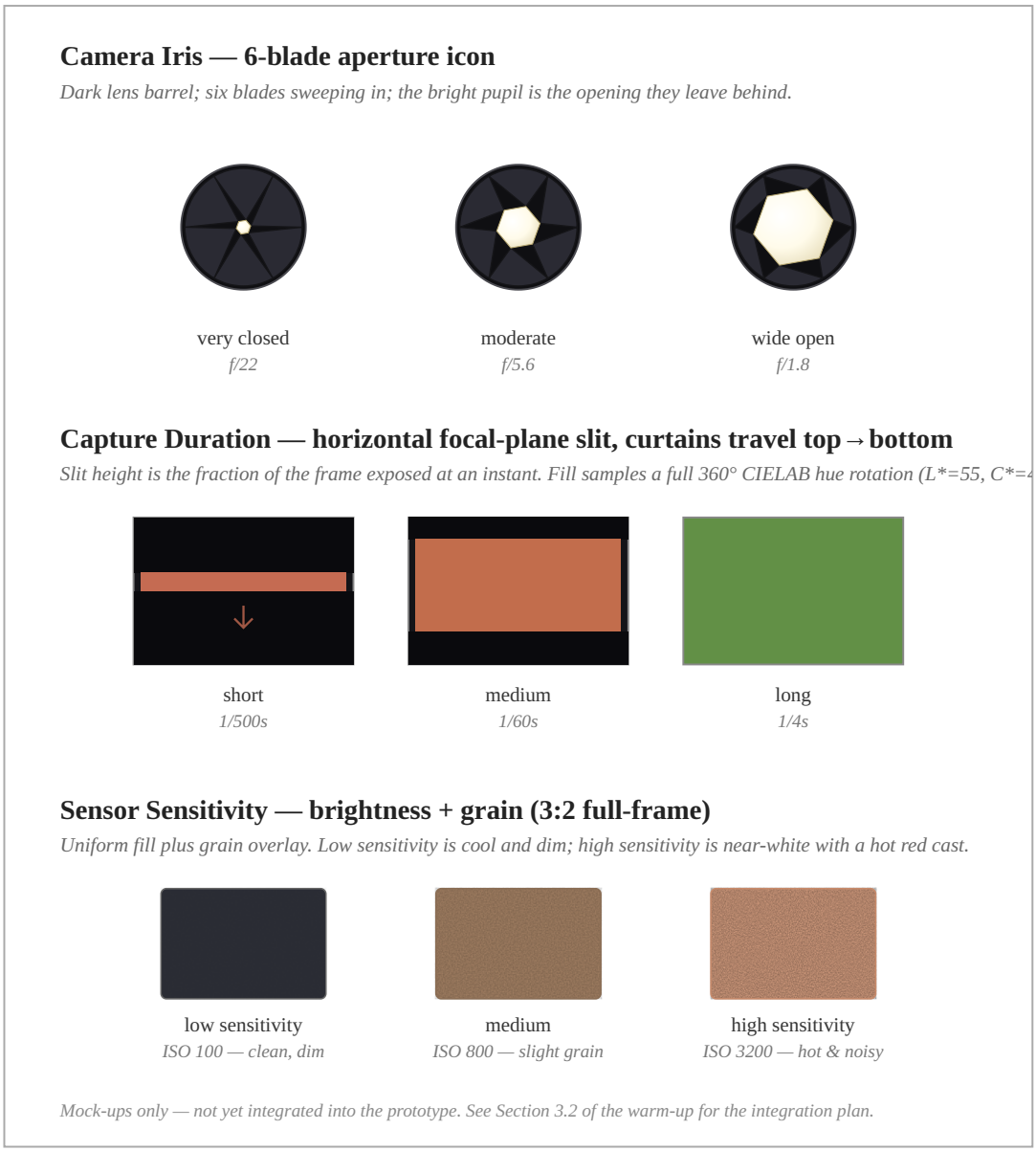
3.2 Difficulty axes (variant dimensions)

1. **Icon scaffolding present / absent.** Icons make each concept visually identifiable; removing them forces reasoning from labels and values alone. Planned replacements for the current placeholder glyphs are purpose-drawn pictorial icons that vary with the *setting value*, previewing the effect inside the icon itself:

    - **Camera Iris / Aperture.** A dark annulus with a bright pupil; pupil size echoes the aperture (tiny pupil for very closed, large pupil for wide open). Reinforces the "your eyes already do this" framing.
    - **Capture Duration / Shutter speed.** A horizontal focal-plane shutter slit on a 3:2 sensor; slit height is the fraction of the frame exposed at an instant. Fill is a full 360° CIELAB hue rotation at one cycle per real second of exposure — sub-second durations barely rotate, so 1/500s and 1/60s read near-identical red and slit geometry carries the distinction; 1/4 s is a quarter turn (green); multi-second exposures cycle through indigo/violet/magenta back to red.
    - **Sensor Sensitivity / ISO.** Uniform fill plus a grain overlay. Low sensitivity is cool and dim; high sensitivity is near-white with a hot red-orange cast, mirroring the

thermal/amplifier-gain character of high-ISO imagery. Grain pattern is the real side effect the setting produces.

Mockups. Working drafts, not yet integrated into the prototype — pulled in here to share current thinking on the visual language. Integration happens alongside the HW10 Flask implementation.



Exposure-setting icon mockups — aperture, capture duration, sensor sensitivity at low / medium / high

A separate file (`shutter_alternatives.svg`) keeps the rotary, focal-plane slit, log-time-bar, and oscilloscope-pulse candidates considered for the capture-duration row, in case the integration work revisits the decision.

2. **Vocabulary scalar / technical.** “Wide open Camera Iris” vs. “f/1.8 aperture.” The lesson teaches scalar; the technical register is the target.

3. **Distractor presence.** An option that matches two of three correct choices but fails on one constraint (as in §2.5 option E).

4. **Irrelevant scenario detail.** Scene properties that do not affect exposure (ambient color, surface reflectivity, subject type) appear in the prompt, testing whether the learner can ignore them.

5. **Constraint count.** Single-constraint (“freeze motion”) vs. multi-constraint (“freeze motion AND avoid grain AND handheld”).

3.3 Why this is HW10-main material, not warm-up material

- Adaptive difficulty needs per-user state (timing, first-try correctness, proficiency estimate, variant selection) that persists across pages.
- That is exactly what the HW10 Flask backend requirement — *“on the backend, you must store important information about users’ choices on every page”* — is designed to support.
- A pure low-fidelity implementation would need either hard-coded branches (not adaptive) or client-side-only state (not persistent). §2.5 is the pre-Flask version; the full system gets built against the Flask backend next week.

4. GitHub repository

- **Repository URL:** https://github.com/rvente/exposure-triangle (public).
- **Contents at submission:**
  - `exposure_triangle.pptx` — PowerPoint export of the Lo-Fi prototype.
  - `README.md` — short description of the project.
- **Collaborators:** the repo is public, so the TA has read access as a GitHub user.

CVN 4170 UI design group Inbox ×

Jett Liuren Wang <jlw2269@columbia.edu> 4:00 AM (17 hours ago)
to Anushka, me, Amina, Nicolas, Nathan, Kennen, Ioannis, Renying, Jane

Hi ALL

I didn't notice until now CVN groups weren't auto assigned, so I thought I'll just send to everyone in CVN and maybe we can form groups here. (list here)

I'm Jett. I finished my undergrad in 2018 CS@UIUC and worked as a UIUX designer in NYC for 5 years before becoming a indie builder & joining CVN (taking classes very slowly). I'm in SF Bay now (PT). It's been nice to get some UX brushup and tie them tighter with code. Happy to join whichever group. Thanks and lmk.

JW

==== Below are my answer to HW6 ====

2. Genshin Impact playable characters.
3. A Genshin Impact player who wants to quickly look up characters and learn about lore and relationships.
4. EX:

```
{
"id": 9,
"name": "Kaeya",
"rarity": 4,
"element": "Cryo",
"img": "https://static.wikia.nocookie.net/gensin-impact/images/2/20/Kaeya_Card.png/revision/latest/scale-to-width-down/1000?cb=20220725205013",
"nation": "Mondstadt",
"affiliation": ["Knights of Favonius", "Khaenri'ah", "Alberich Clan", "Ragnvindr Clan"],
"story": "In the Knights of Favonius, Kaeya is the most trusted aide for the Acting Grand Master Jean. You can always count on him to solve any intractable problems. Everyone in Mondstadt loves Kaeya, but no one knows what secrets this witty, charming knight has...",
"wiki": "https://genshin-impact.fandom.com/wiki/Kaeya"
}
```

Ioannis N Daras 5:07 PM (4 hours ago)
Hi everyone, Here are my answers for HW6: For my dataset, I'm building "Best Mystery Movies After 2010." The site will allow users to find mystery and psycholog

Blake Vente 9:57 PM (0 minutes ago)
to Ioannis, Jett, Anushka, Amina, Nicolas, Nathan, Kennen, Renying, Jane

Hey all, nice to meet you.

I'm a software engineer at an audio media company headquartered in Newark. Also taking classes very slowly. This course has been changing the way I see so many of the user experience flows around me.

What data will your website allow users to search?

Philosophically immersive fiction — books, TV shows, and films where philosophical questions are the engine of the narrative rather than set dressing. I'm calling the collection "Philo-core" internally, but the site will present it as a curated database of fiction that rewards deep thinking.

Who is an example user, and why would they want to search this?

A fan of Hard Science fiction who just finished Martha Wells' Murderbot Diaries and was struck by how deeply the series explores questions of autonomy, personhood, and what it means to choose your own life when you were built to serve. They want to find other fiction — across books, TV, and film — where philosophical questions aren't just backdrop but are the actual driving force of the story. They want to filter by specific philosophical themes (e.g., consciousness, identity, free will, ethics of AI, moral philosophy) and compare works by how philosophically substantive they are, so they can prioritize deeply immersive works over ones that merely gesture at ideas. They also want to discover entire series once they find a single entry they like, and to understand *why* each work was included in the collection. This user might have realized that they can gain new rich philosophical insights from attending to stories where there are rich stories that make the concepts themselves memorable.

Hope I didn't make you all read too much.

Best,
Blake

Screenshot of the repository name and members

5. Job assignments (solo CVN adaptation)

- CVN student with no formal project group; all roles fall to one person.
- Per HW7 precedent (*"As a CVN student, I do not have a formal project group"*) and the HW10 rubric's note that six roles can overlap when the group is smaller than six.

Part	Role	Assigned to
Learning portion	Architecting the data	Blake Vente (rv2459)

Part	Role	Assigned to
Learning portion	Implementing the UI	Blake Vente (rv2459)
Learning portion	Testing that you can click through the app	Blake Vente (rv2459)
Quiz portion	Architecting the data	Blake Vente (rv2459)
Quiz portion	Implementing the UI	Blake Vente (rv2459)
Quiz portion	Testing that you can click through the app	Blake Vente (rv2459)

- Work is self-sequenced: data schema and JSON fixtures first (shared by both portions), then Flask routes and templates, then quiz-state persistence, with click-through testing interleaved per route.

6. HW10 main-assignment implementation plan (next week)

- **Backend:** Flask + SQLite (simplest persistent store for the single-user assumption).
- **Routes:** `/` (home + start), `/learn/<n>` (lesson slides), `/quiz/<n>` (quiz questions), `/result` (quiz summary). ~4 routes as the rubric suggests.
- **Data:** lessons and quiz questions in JSON fixtures; routes render each page from data so content changes do not require template edits.
- **Per-page state:**
  - Lessons — entry time, slider/toggle values settled on, time spent.
  - Quizzes — attempt history (first-try correctness, final answer, latency), feeding the adaptive-difficulty scoring.
- **Scoring:** each question weighted by inverse difficulty (inverse of historical correct-rate × response-time). In single-user mode reduces to per-question constants, but the schema is designed so multi-user data would drive weights.
- **Component library:** interaction primitives surfaced in the low-fi prototype (snap-on-release slider, discrete toggle group, image-choice buttons, comparison slider, MC with second-chance) become a shared JS/CSS component set so new lessons can be authored quickly.